

Final Report

Study of Massive and Massless Particle Trajectories Near Schwarzschild and Kerr Black Hole

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1 Introduction:

This fellowship is focused on building a working understanding of **general relativity** through *computational methods*, using black holes as the primary system to study. Since the equations describing how particles and light move in curved spacetime are difficult to solve exactly, numerical methods were used to compute these trajectories. The project covers two black hole spacetime, the **Schwarzschild** black hole and the **Kerr** black hole, starting with computing and validating particle and photon trajectories around each. For the Schwarzschild case, this was extended further to study how the impact parameter and energy of photons detected far from the black hole depend on the motion of the source emitting them and the position of the observer, using photons emitted from the **innermost stable circular orbit** (ISCO).

2 Theoretical Background:

This project required a few prerequisites in general relativity, namely *spacetime geometry* and the effects of *mass* and *speed* on it. Furthermore, *numerical methods* were used for the simulations.

2.1 Space-time geometry:

- **Metric and Christoffel symbols -**

To understand spacetime, we use coordinate systems, and to measure quantities within these coordinate systems we use a *metric*. Just as we need a measuring tape to measure distances, we need a metric to measure distances in spacetime. In this project I have dealt with Schwarzschild and Kerr metric specifically.

$$\text{Schwarzschild Metric } (g_{\mu\nu}) = \begin{bmatrix} -(1 - \frac{2M}{R}) & 0 & 0 & 0 \\ 0 & (1 - \frac{2M}{R})^{-1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{bmatrix}$$

Christoffel symbols describe how the coordinate basis vectors change from point to point, which is what allows us to differentiate tensors consistently on a *curved manifold*.

$$\Gamma^\alpha_{\mu\nu} = \frac{1}{2} g^{\alpha\beta} (g_{\beta\mu,\nu} + g_{\beta\nu,\mu} - g_{\mu\nu,\beta}) \quad (1)$$

- **Riemann Tensor:**

The Riemann tensor measures the curvature of spacetime itself. Specifically, how a vector changes when it is *parallel transported* around a small closed loop. If spacetime is flat, this change is zero; if spacetime is curved, the vector returns rotated, and the Riemann tensor quantifies exactly how much.

$$R^\alpha{}_{\beta\mu\nu} = \Gamma^\alpha{}_{\beta\nu,\mu} - \Gamma^\alpha{}_{\beta\mu,\nu} + (\Gamma^\alpha{}_{\sigma\mu}\Gamma^\sigma{}_{\beta\nu}) - (\Gamma^\alpha{}_{\sigma\nu}\Gamma^\sigma{}_{\beta\mu}) \quad (2)$$

- **Ricci tensor and Ricci scalar:**

Obtained by *contracting* the Riemann tensor, condensing its full curvature information into a simpler quantity that captures how volumes change due to curvature. This simplified form is what actually appears in the *Einstein field equations*, linking spacetime curvature to the matter and energy present.

- **Geodesic equation:**

The geodesic equation describes the path a *free-falling* particle or photon follows through curved spacetime. It is derived by requiring that the particle's **four-velocity** be parallel transported along its own path, and the Christoffel symbols appear directly in this equation to account for the curvature of the coordinate system.

$$\frac{d^2 x^\alpha}{d\lambda^2} + \Gamma^\alpha{}_{\mu\beta} \frac{dx^\mu}{d\lambda} \frac{dx^\beta}{d\lambda} = 0 \quad (3)$$

2.2 Effect of Mass on Spacetime Geometry:

- **Einstein Tensor:**

The Einstein tensor is built from the Ricci tensor and Ricci scalar, and it captures how spacetime curvature relates to the distribution of mass and energy. It forms the left-hand side of the Einstein field equations, which state that spacetime curvature is directly proportional to the matter and energy present given by the stress-energy tensor.

$$R_{\alpha\beta} - \frac{1}{2}g^{\alpha\beta}R = 8\pi T^{\alpha\beta} \quad (4)$$

2.3 Effect of Speed on Spacetime Geometry:

- **Aberration:**

Aberration is the special relativistic effect where the observed direction of an incoming light ray depends on the relative velocity between the source and the observer, causing angles to appear shifted compared to the non-relativistic case. This becomes important when comparing detection angles for a stationary versus a moving source in your ISCO study, since the source's motion shifts where the observer perceives the photon to be coming from.

The aberrated angle θ is given by:

$$\cos(\theta) = \frac{\cos(\theta_o) - \beta}{1 - \beta\cos(\theta_o)} \quad \text{and} \quad \sin(\theta) = \frac{\sin(\theta_o)}{\gamma(1 - \beta\cos(\theta_o))} \quad (5)$$

2.4 Numerical Method:

- **Runge-Kutta Method:**

RK-4 method is a numerical analysis method that helps to predict the following values of a function based on the initial conditions of the system or the initial values of the variables of the function. I have used this method to find the successive values of the non-linear differential equations that are used.

$$RK4 : y_{n+1} = y_n + hT_4(x_n, y_n, h) \quad (6)$$

where;

$$T_4 = \frac{(k_1 + 2k_2 + 2k_3 + k_4)}{6} \quad (7)$$

3 Schwarzschild: Trajectories and Validation

I have simulated the trajectories of massive and massless particles around a Schwarzschild black hole using the geodesic equations and the RK-4 method. This includes checking the validity of the simulated data by comparing it against the weak-field approximation. I have further extended this approach to the Kerr black hole.

3.1 Computing Trajectories:

All of the simulations consider that the events happening are in the equatorial plane of the black hole, meaning $\theta = 90$ degrees.

3.1.1 Equations used;

- For massless particles:

$$\left(\frac{dr}{d\lambda}\right)^2 = E^2 - \left(1 - \frac{R}{r}\right) \left(\frac{L^2}{r^2}\right) \quad \text{and} \quad \frac{d\phi}{d\lambda} = \frac{L}{r^2} \quad (8)$$

where,

- Spherical coordinates - radial distance(r), azimuthal angle(ϕ)
- Total energy - E
- Schwarzschild radius - R
- Angular momentum - L
- Affine parameter - λ

- For massive particles:

$$\left(\frac{dr}{d\lambda}\right)^2 = E^2 - \left(1 - \frac{R}{r}\right) \left(\frac{L^2}{r^2}\right) - \epsilon \left(1 - \frac{R}{r}\right) \quad \text{and} \quad \frac{d\phi}{d\lambda} = \frac{L}{r^2} \quad (9)$$

here, $\epsilon = 1$ for massive particles while it is zero for massless particles.

- Weak-field approximation:

$$\Delta\phi = \frac{2R}{b} = \frac{4GM}{bc^2} \quad (10)$$

where,

- Impact parameter - b
- Gravitational constant - G
- Mass of black hole - M

3.1.2 Computational Results:

Since I am dealing with squared of derivative term ($dr/d\lambda$), I differentiate them again to obtain a simpler second-order ordinary differential equation.

$$\frac{d^2r}{d\lambda^2} = \frac{L^2}{r^3} - \frac{3}{2} \frac{RL^2}{r^4} - \frac{\epsilon R}{2r^2} \quad (11)$$

Since I was using the RK-4 method, there were some initial conditions that were needed to be assigned for computing the trajectories. The radial distance(r) and the azimuthal angle(ϕ) were hard coded while the derivative $dr/d\lambda$ has to be measured via;

$$\frac{dr}{d\lambda} = \sqrt{E^2 - \left(1 - \frac{R}{r}\right) \frac{L^2}{r^2} - \epsilon \left(1 - \frac{R}{r}\right)} \quad (12)$$

- **Massless Particle:**

One of the most interesting phenomena observed was existence of **photon sphere**. A photon sphere is a ring around a black hole where the photons are trapped in an orbit due to the gravity of the black hole. This means, photons can undergo unstable circular orbits near the photon sphere which is **1.5** times the *Schwarzschild radius*(R) of the black hole away from its centre.

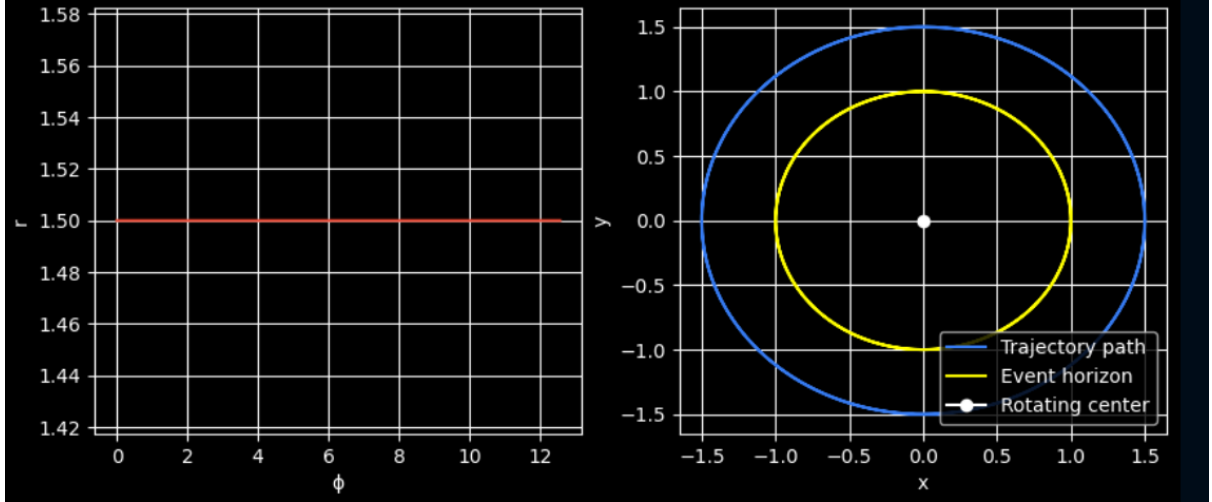


Figure 1: Photon sphere

- **Massive Particle:**

Another interesting phenomenon observed in trajectories around the black hole was **perihelion shift**. Here, a massive particle such as a star or planet revolving around the black hole in a bound orbit experiences a gradual shift in the orientation of its orbit over time, rather than retracing the same closed ellipse each revolution as predicted by *Newtonian gravity*. This shift arises because the geodesic equations in curved spacetime do not produce perfectly closed elliptical orbits, and the perihelion precesses slightly with each orbit.

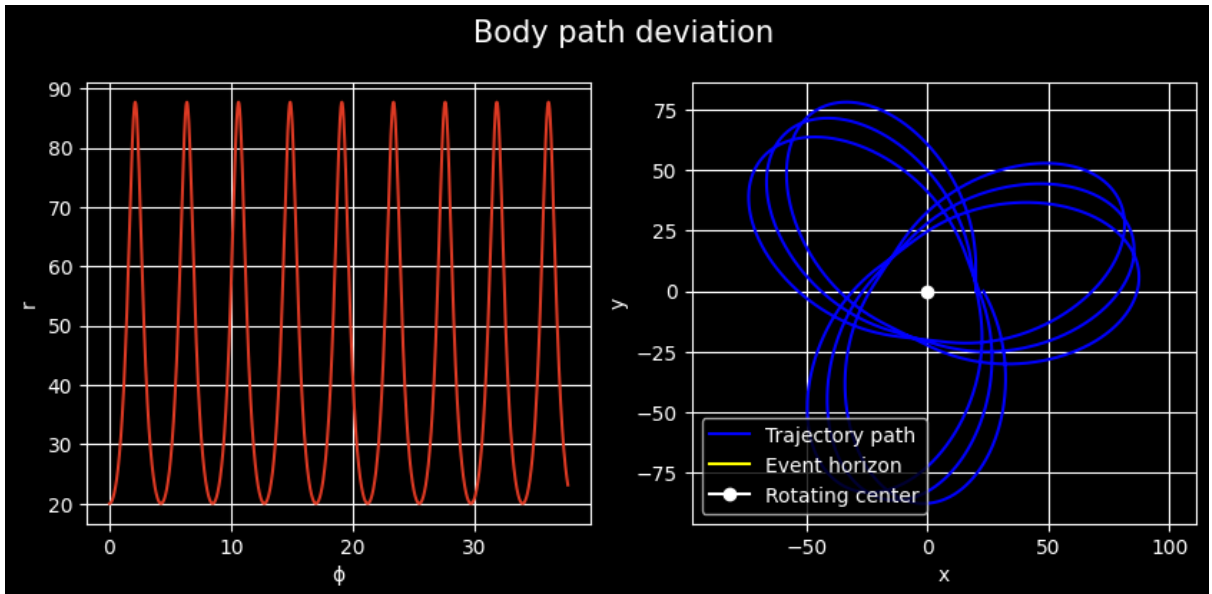


Figure 2: Perihelion shift

3.1.3 Numerical Analysis:

To verify the correctness of the simulated trajectories, the results were checked against the weak-field approximation an analytical expression for the deflection angle that is valid when the impact parameter is large compared to the Schwarzschild radius. In this, the deflection angle predicted by general relativity is known in closed form, and the simulated trajectories should agree closely with this prediction.

This was checked by plotting the ratio $\Delta\phi_{sim}/\Delta\phi_{weak}$ against the impact parameter. For large impact parameters, this ratio approaches **1**, confirming that the simulation correctly reproduces the analytical result in the regime where the approximation is expected to hold. As the impact parameter decreases and the trajectory approaches the black hole, the ratio deviates from 1, reflecting the fact that the weak-field approximation breaks down in the strong-field, which is expected, since the approximation is not valid there in the first place.

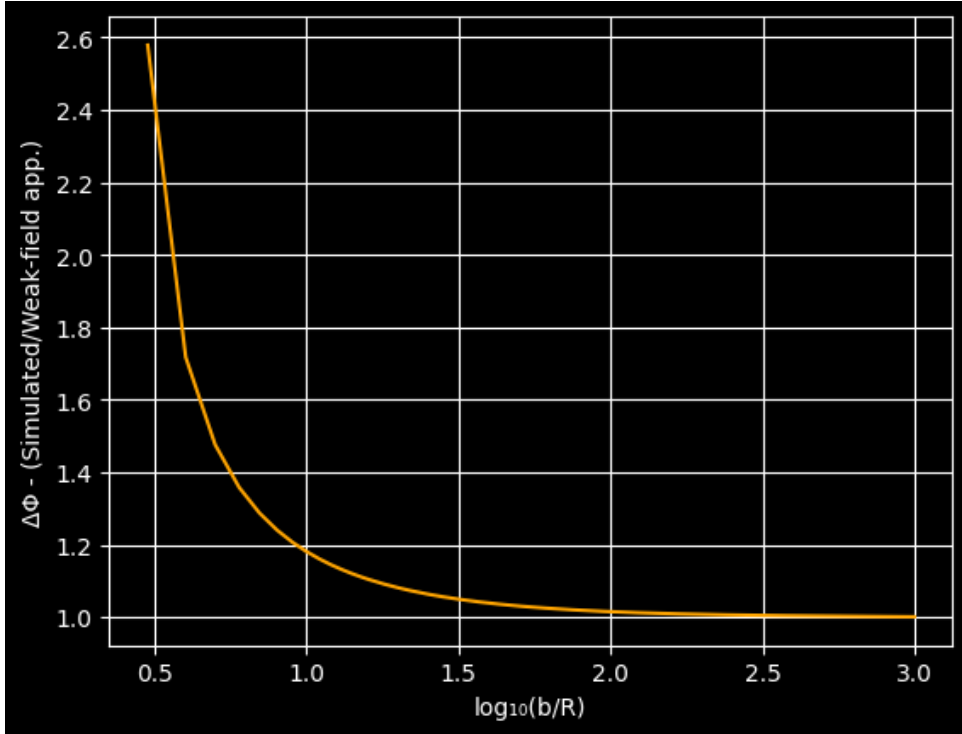


Figure 3: Simulated data and weak field approximation comparison

4 Kerr: Trajectories and Validation

As mentioned earlier, the idea of simulating trajectories and validating them was extended to the Kerr black hole as well. Unlike the Schwarzschild black hole, which is non-rotating, a Kerr black hole **rotates** about its own axis. This rotation has several consequences and gives rise to a number of new phenomena, including spherical asymmetry, a ring singularity, frame dragging, and the existence of an ergosphere.

- Spherical asymmetry -

Firstly, the rotation breaks the spherical symmetry observed for the Schwarzschild black hole. A Kerr black hole is not spherically round but **oblate**, bulging outward at the equator due to the rotation.

- Ring singularity -

The most common coordinate system used to describe the Kerr metric is the *Boyer-Lindquist coordinate system*. In these coordinates, the singularity is no longer a single point (as in the Schwarzschild case) but takes the shape of a ring, referred to as a **ring singularity**, or "**ringularity**."

- Frame Dragging -

Since the Kerr black hole is rotating, it drags the surrounding spacetime along with it in the direction of its rotation, an effect known as **frame dragging**. This is a direct consequence of the rotation and is the underlying cause of the *ergosphere*.

- Ergosphere -

The ergosphere is a region surrounding a rotating black hole, outside the event horizon, within which no particle can remain stationary; all particles are forced to co-rotate with the black hole due to frame dragging. This effect applies even to photons. A photon traveling opposite to the black hole's rotation, upon approaching the ergosphere, is gradually slowed in its counter-rotation and effectively becomes momentarily "trapped" at the boundary before being dragged along in the direction of rotation.

4.1 Equations used:

- **Kerr metric from Boyer-Lindquist coordinates:**

$$g_{\mu\nu} = \begin{bmatrix} -\left(1 - \frac{2\mu r}{\Sigma}\right) & 0 & 0 & -\left(\frac{2\mu a r \sin^2 \theta}{\Sigma}\right) \\ 0 & \frac{\Sigma}{\Delta} & 0 & 0 \\ 0 & 0 & \Sigma & 0 \\ -\left(\frac{2\mu a r \sin^2 \theta}{\Sigma}\right) & 0 & 0 & \sin^2 \theta \left(r^2 + a^2 + \frac{2a^2 \mu r \sin^2 \theta}{\Sigma}\right) \end{bmatrix} \quad (13)$$

where,

$$\Sigma = r^2 + a^2 \cos^2 \theta$$

$$\Delta = r^2 - 2\mu r + a^2$$

$$\mu = GM/c^2$$

a = J/Mc, a - specific angular momentum, J - total angular momentum

- **Geodesic Equations:**

$$\left(\frac{dr}{d\lambda}\right)^2 = E^2 + \frac{a^2(E^2 + \epsilon) - L^2}{r^2} + \frac{R(L - aE)^2}{r^3} - \epsilon \left(1 - \frac{R}{r}\right) \quad (14)$$

$$\frac{d\phi}{d\lambda} = \frac{1}{\Delta} \left(\frac{ERa}{r} + L \left(1 - \frac{R}{r}\right) \right) \quad (15)$$

$$R = 2GM/c^2 = 2\mu$$

4.2 Computational Results:

4.2.1 Trajectories:

There are two cases observed in the case of a rotating black hole;

- **Prograde:** where the particle and the black hole is spinning in **same** direction. A prograde photon gets dragged along with spacetime which causes it to linger around the black hole for much longer periods.

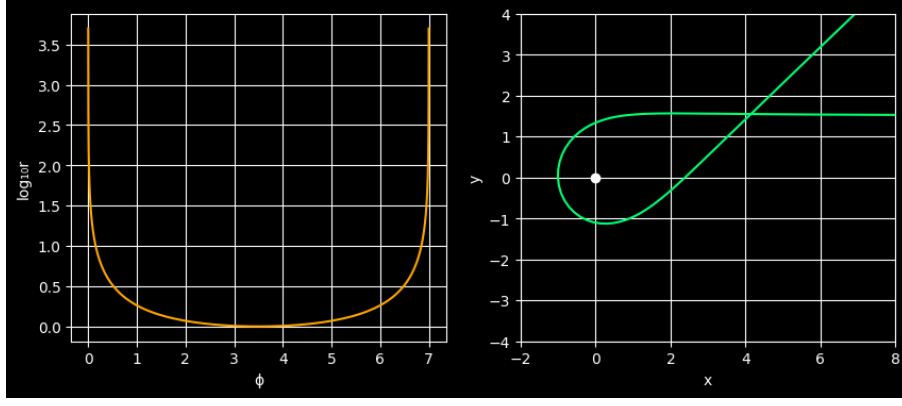


Figure 4: Prograde photon

- **Retrograde:** where the photon and the black hole is spinning in **opposite** direction. This leads to an interesting observation, the photon is eventually forced to stop and follow the direction of the rotation of the black hole.

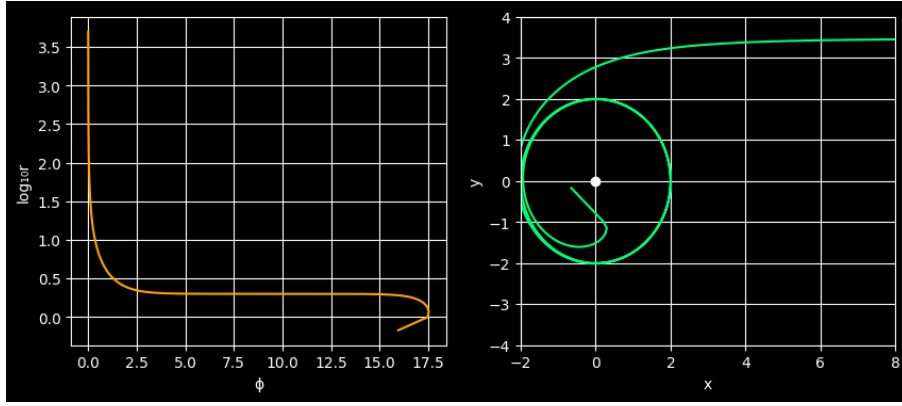


Figure 5: Retrograde

Moreover, here is a comparison for prograde and retrograde trajectory of a photon for the same impact parameter ($\mathbf{b} = \mathbf{3.5}$).

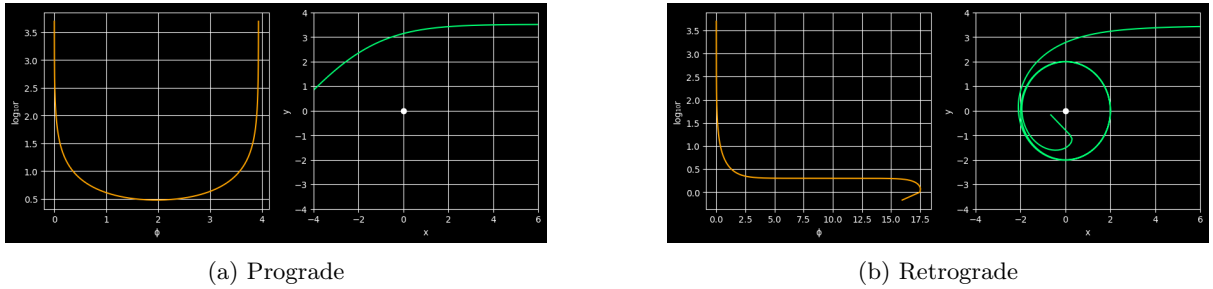


Figure 6: Trajectory of photon for Kerr black hole

4.2.2 Validation:

As with the Schwarzschild case, the simulated Kerr trajectories were checked against the weak-field approximation by plotting the ratio $\Delta\phi_{sim}/\Delta\phi_{weak}$ against the impact parameter. However, since the weak-field approximation used here does not account for rotation, it predicts a single deflection angle for a given impact parameter regardless of orbital direction, whereas the simulated data splits into two distinct trends for prograde and retrograde orbits.

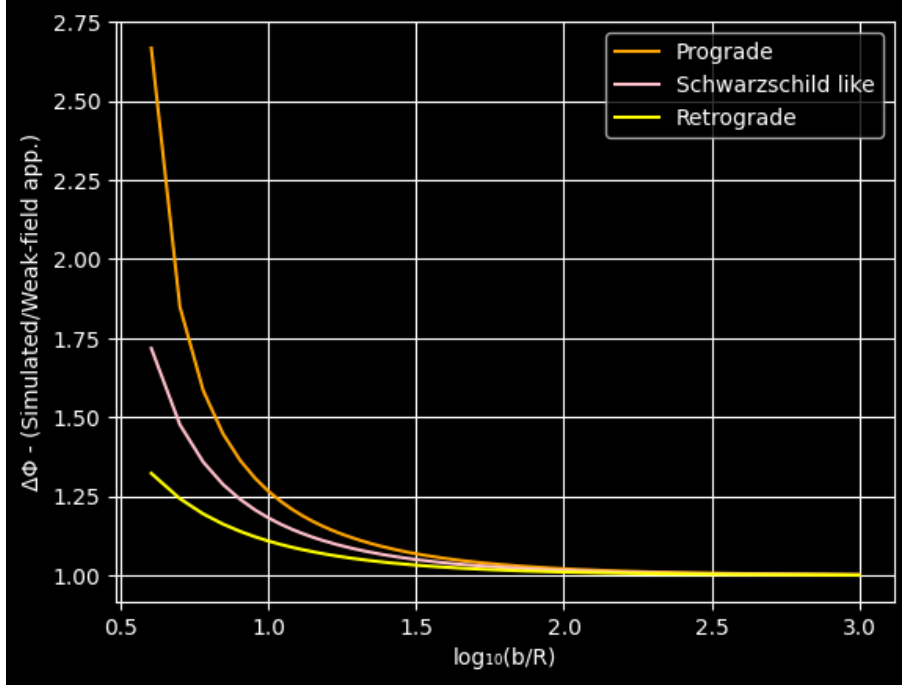


Figure 7: $\Delta\phi_{sim}/\Delta\phi_{weak}$

For large impact parameters, both prograde and retrograde ratios approach 1, confirming agreement with the non-rotating weak-field limit where frame-dragging effects become negligible. As the impact parameter decreases, however, the two trends **diverge** from one another and from the weak-field prediction. The prograde ratio deviating in one direction and the retrograde ratio in the other.

This reflects the fact that the weak-field approximation cannot capture the direction dependent effect of frame dragging that becomes significant closer to the black hole.

5 Photon Detection from the ISCO:

The next phase consisted of simulating multiple trajectories of light rays emitted from a source orbiting a Schwarzschild black hole. Black holes have stable orbits in which a particle remains bound, neither escaping to infinity nor falling into the black hole over time. The Schwarzschild black hole has an ISCO, which is the closest such stable orbit to the black hole, any circular orbit nearer than this is unstable, and a small perturbation will cause the particle to spiral inward. For massive particles, the ISCO radius is $r_{ISCO} = 3R_s$, where R_s is the Schwarzschild radius of the black hole.

5.1 Theoretical Derivation:

Deriving the radius of the orbit using:

$$\frac{dr}{d\lambda} = 0, \quad \text{substituting in; } E^2 = \left(\frac{dr}{d\lambda}\right)^2 - \frac{L^2}{r^2} \left(1 - \frac{R}{r}\right) \quad (16)$$

Differentiating with respect to r:

$$\frac{R}{r^2} - \frac{2L^2}{r^3} + \frac{3RL^2}{r^4} = 0 \quad (17)$$

Since total energy(E) is constant, its derivative equals to zero. We derivate the above equation and finally solve for:

$$r^2 - 3L^2r + L^2 = 0 \quad (18)$$

Where we obtain an expression for L^2 from (2) and lastly we obtain the ISCO radius:

$$r_{ISCO} = 3R = \frac{6GM}{c^2} \quad (19)$$

For computing trajectories the formula remains the same:

$$\frac{dr^2}{d\lambda} = E^2 - \left(1 - \frac{R}{r}\right)\left(\frac{L^2}{r^2}\right) \quad (20)$$

5.2 Computational Results:

I divided this phase into three parts, starting with a stationary observer at $(r, \phi) = (100, 180)$ and a stationary source at the ISCO, $(r, \phi) = (3R, 0)$, from which **36** and **72** rays with equal angular interval were simulated. Since the simulated rays rarely intersected the observer's exact position, the observer was instead treated as a circular detection region of radius **5R**. The impact parameters of the detected rays were then plotted, along with their energies, computed via gravitational redshift, assuming an incident frequency of **1 Hz**.

$$v_{obs} = v_{emit} \sqrt{\frac{1 - \frac{R}{r_0}}{1 - \frac{R}{r_{obs}}}} \quad (21)$$

5.2.1 Part I: Stationary observer and stationary source

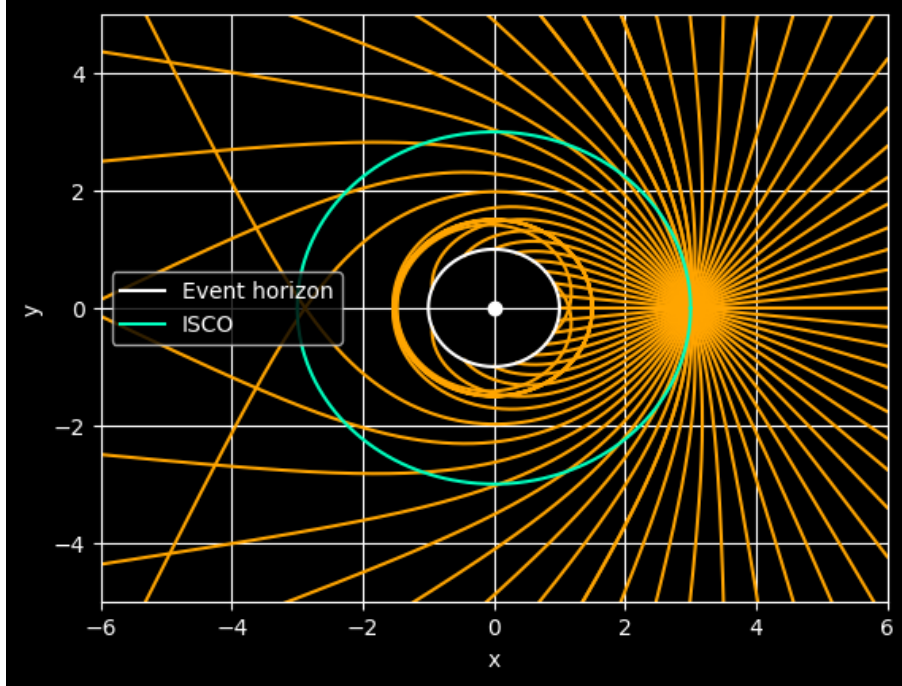


Figure 8: 72 rays from stationary source in ISCO

- In this 2-D plot these images that are formed look like dots of light, but when extended to third dimension, these images form a ring-like image of light from the source called *Einstein's ring*.
- For **smaller** intervals **more** number of images are formed.
- For an image to be formed the impact parameter for the trajectory of that photon should be greater than **critical impact parameter** (b_{crit}) where;

$$b_{crit} = \frac{3\sqrt{3}R}{2} = \frac{3\sqrt{3}(1)}{2} = 2.598 \quad (22)$$

- This b_{crit} is what determines the diameter of the Einstein ring that is formed. The diameter depends on the Schwarzschild radius of the black hole and ultimately the mass(M) of the black hole.

5.2.2 Part II: Stationary observer and moving source

In this part, the source moves along the orbit, and so the special-relativistic Doppler effect and aberration along with gravitational redshift must be taken into account. Using these effects, a graph was plotted showing the ratio of the energy received by the observer to the source's emitted energy, against the angular position of the source. The tangential velocity of the source used in this simulation was derived theoretically, as shown below.

Derivation:

For a circular orbit:

$$\frac{dr}{d\lambda} = 0, \quad \text{substituting in; } E^2 = \left(\frac{dr}{d\lambda}\right)^2 - \left(\frac{L^2}{r^2} + \epsilon\right) \left(1 - \frac{R}{r}\right) \quad (23)$$

On solving for E and L: I get $E = \frac{2\sqrt{2}}{3}$ and $L = \sqrt{3}$.

We know;

$$\frac{d\phi}{d\lambda} = \frac{L}{r^2}, \quad \text{and} \quad \frac{dt}{d\lambda} = \frac{E}{\left(1 - \frac{R}{r}\right)} \quad (24)$$

We can find the tangential velocity for the source by;

$$v_{tangent} = r\sqrt{g_{rr}}\frac{d\phi}{dt} = \left(1 - \frac{R}{r}\right)^{-1/2} \frac{L \left(1 - \frac{R}{r}\right)}{Er} \quad (25)$$

Hence,

$$v_{tangent} = \frac{L\sqrt{1 - \frac{R}{r}}}{Er} = 0.5 \quad (26)$$

Lastly,

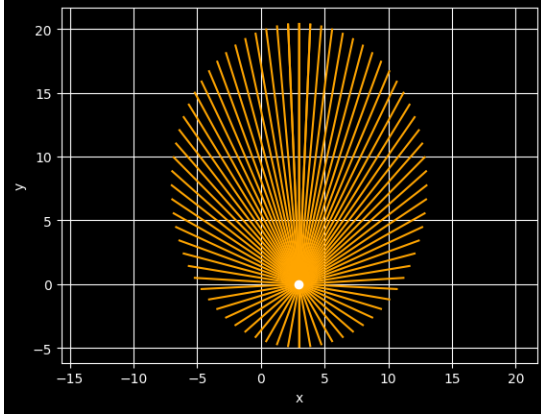
$$\beta = \frac{v}{c} \quad \text{and} \quad \gamma = \frac{1}{\sqrt{1 - \beta^2}} \quad (27)$$

Considering the gravitational redshift and Doppler effect, the total redshift that we get is:

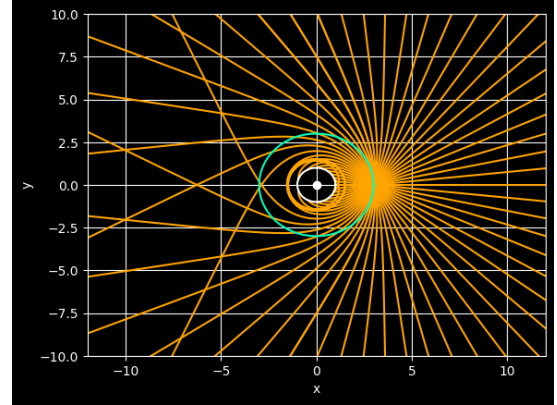
$$\nu_{obs} = \nu_{emit} \left(\frac{1 - \beta \cos(\theta_o)}{\sqrt{1 - \beta^2}} \right) \sqrt{\frac{1 - \frac{R}{r_o}}{1 - \frac{R}{r_{obs}}}} \quad (28)$$

$$E_{obs} = E_{emit} \left(\frac{1 - \beta \cos(\theta_o)}{\sqrt{1 - \beta^2}} \right) \sqrt{\frac{1 - \frac{R}{r_o}}{1 - \frac{R}{r_{obs}}}} \quad (29)$$

Computational Results:



(a) Ray-length as a function of aberrated energy



(b) ISCO simulation with gravity considered

Figure 9: Trajectory of (72)light rays emitted anti-clockwise moving source

Observations:

Using the similar set-up of an observer being placed at $(\mathbf{x}, \mathbf{y}) = (-100, 0)$, I have obtained the *impact parameters*(b) and the *energies*(E) of the rays that are being detected.

Since the rays rarely hit the $(x, y)_{obs}$ directly, I have considered the observer to be a circular region of radius $(7.5)\mathbf{R}$.

5.2.3 Impact parameter vs ϕ :

This is the plot for the impact parameter(b) of the detected rays for a moving source at different ϕ angles.

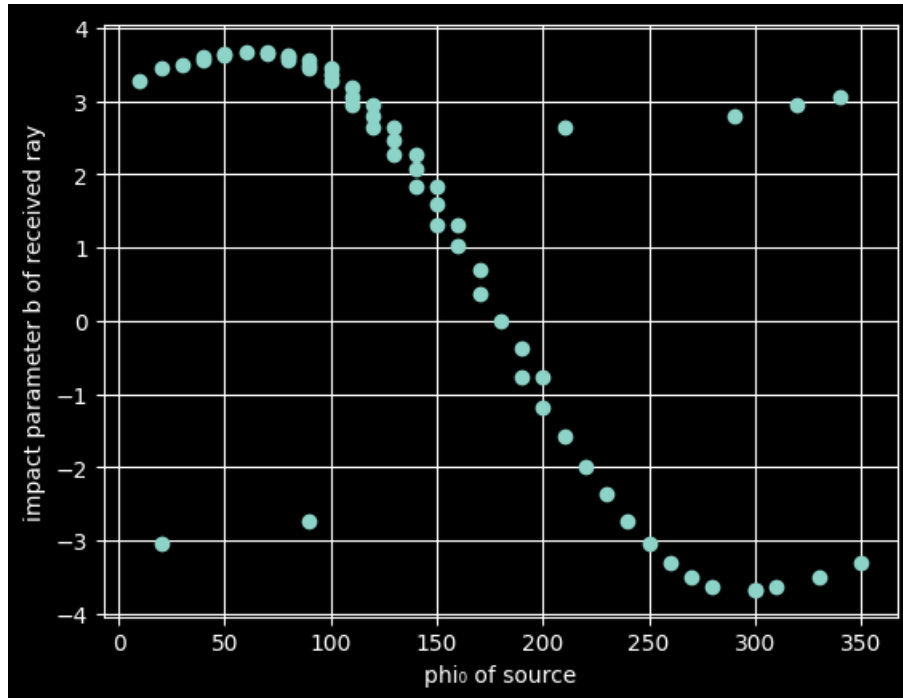


Figure 10: Impact parameter(b) vs angular-position(ϕ)

- The graph is smooth and continuous but yields a few discrepancies. As observed there are a few points nowhere near the smooth curve.
- These points are not **errors**. These impact parameters as seen from the graph are close to the *critical impact parameter* ($b_{crit} = 2.6$) value. Any impact parameter lower than this causes the photon to sink in the black hole.
- At these impact parameters the photon travels close enough to not get pulled in by the black hole and the deviation is so high that it hits the observer at **(-100,0)**.

5.2.4 Energy vs ϕ :

Similarly, this is a plot for the energies(E) of the photons that are being detected as a function of the position(ϕ) of the moving source. The frequency emitted by the source is **1Hz** and ($h=1$) in *natural units*, $E = h\nu_{emit} = 1 \cdot 1 = 1s^{-1}$.

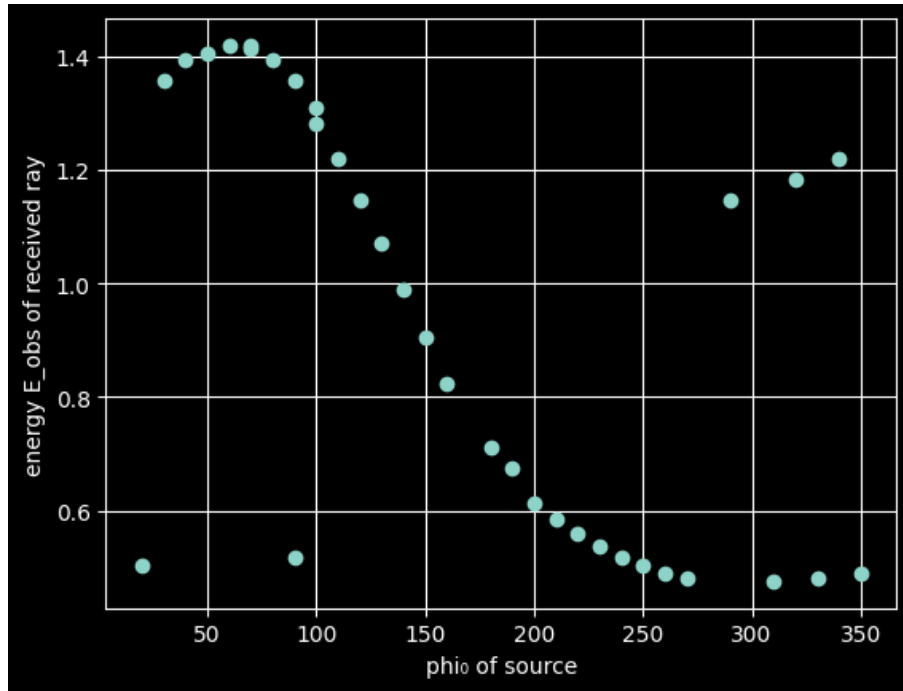


Figure 11: Energy detected(E) vs angular-position(ϕ)

Observations:

- Similar to the impact parameters vs ϕ graph we have some anomalies in the trend. As seen there are two rays getting suddenly redshifted around **50** and **100** while a few get unexpectedly blueshifted at **300** and **350**.

- **The reason for unexpected redshift;**

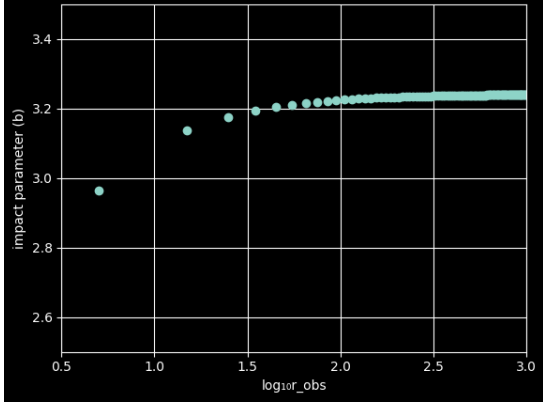
Referring back to the impact parameter graph, and focusing specifically on the out-of-curve values, it can be seen that the photons detected with unexpectedly high redshift are those emitted in the direction opposite to the source's motion. This additional redshift arises from the Doppler effect, which compounds with the gravitational redshift already present.

- **The reason for unexpected blueshift;**

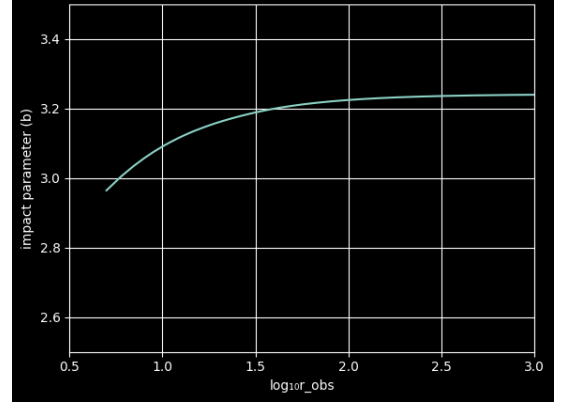
Similarly, the impact parameter graph shows that the photons responsible for the unexpected blueshift are emitted in the direction of the source's motion, which causes them to be Doppler-blueshifted at emission. Despite this, the net observed effect remains a blueshift because, at the observer's distance, the Doppler blueshift outweighs the gravitational redshift that would otherwise dominate.

5.2.5 Part III: Moving observer and moving source

In this section, the impact parameters of the detected rays are analyzed as a function of the observer's position, divided into two sub-parts: (a) varying radial distance(r), and (b) varying angular position(ϕ). The detections are made when the source is at $(r, \phi) = (3R, 0)$.

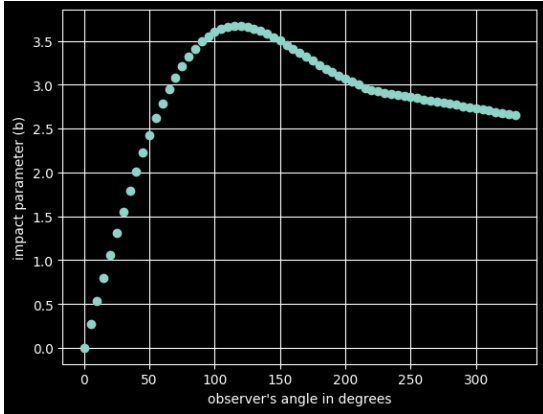


(a) Scattered graph

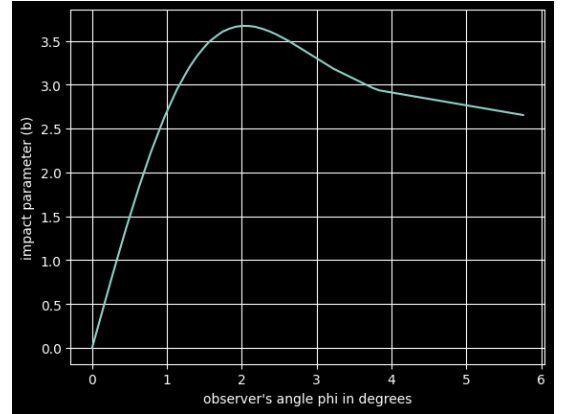


(b) Interpolated graph

Figure 12: impact parameter (b) vs radial distance of observer (r_{obs})



(a) Scattered graph



(b) Interpolated graph

Figure 13: impact parameter (b) vs angular-position of observer (ϕ_{obs})

6 Conclusion:

This fellowship provided a working understanding of general relativity, from its theoretical foundations to their computational implementation. Beginning with spacetime geometry — the metric, Christoffel symbols, and the Riemann, Ricci, and Einstein tensors — the geodesic equation was derived and numerically integrated using the RK-4 method to simulate trajectories of massive and massless particles around a Schwarzschild black hole. These simulations were validated against the weak-field approximation and shown to correctly reproduce known effects such as perihelion shift.

This framework was then extended to the Kerr black hole, where rotational effects — spherical asymmetry, frame dragging, the ergosphere, and the ring singularity — introduced complexity absent in the Schwarzschild case. Weak-field validation here revealed a clear prograde-retrograde asymmetry due to frame dragging, confirming that the simulation correctly captures rotational effects. Due to the instability of the Kerr ISCO, the ray-detection studies used for Schwarzschild could not be directly extended to the rotating case.

The final phase studied photons emitted from the Schwarzschild ISCO, progressing from a stationary source and observer, to a moving source with Doppler and aberration effects, and finally to a varying observer position. This showed that the impact parameter and energy of detected photons depend on both gravitational effects and the relative motion and position of the source and observer, with Doppler shift and gravitational redshift interacting in occasionally counter-intuitive ways.

Overall, this project moved from the mathematical foundations of general relativity to concrete, physically interpretable results for both static and rotating black holes, establishing a solid computational base for future work in relativistic astrophysics.